

Q. How does one avoid latency in edge deployments?

A. Contrary to popular belief, you don't always need a fast connection at the edge. While that's sometimes true, in many cases, it's not. We've found that if you design the network correctly, with a failure domain where most processes can run independently, latency becomes less of an issue and is manageable. One of our customers operates in an environment where edge locations are connected via satellite, which has high latency and low bandwidth. As a result, they've adapted their CI/CD system so that major updates are scheduled when these mobile edge 'locations' are in an area with stronger and faster connectivity, while smaller updates can still be pushed at other times.

Q. How is security at the edge handled in case of Kubernetes clusters?

A. Security strategy needs to evolve when pushing compute and

storage resources to the edge, as these resources are no longer in a hardened data centre but in more vulnerable locations, where there is physical access to the machines. The most effective approach is adopting a zero trust security model, where you assume the environment is hostile from the start. Every connection requires mutual identification processes, embedded via protocols such as mutual TLS (mTLS).

SUSE NeuVector is based on zero trust principles and covers three to four of the core zero trust pillars outlined by the Zero Trust Reference Architecture published by security agencies like DISA. It offers container isolation and secures the Kubernetes cluster, while complementing other security measures like identity management and monitoring to round out a complete security strategy at the edge.

Q. How do you ensure that all these deployments meet

regulatory and compliance standards?

A. We identify the risk management framework that applies to our specific situation, and then conduct an assessment against our default system to identify how it meets the required technical controls. This process involves checking off security measures for the Linux layer, the Kubernetes layer, and providing guidance for the application layer. After confirming these measures are in place, we codify them into a policy and implement it at the edge to ensure compliance with the entire framework.

While I have simplified this process, the overarching steps are clear: identify compliance needs, perform an assessment, push the findings into production, and continuously reassess and audit the security state. This includes ongoing monitoring to adapt to any policy changes or emerging threats. We assist with this entire process to ensure compliance at every stage. **END** 

OSFY Magazine Attractions During 2024-25

Month	Theme
March 2024	Gen AI, Deep learning and Machine Learning
April 2024	Open Source Programming (Languages and tools)
May 2024	Blockchain and Open Source
June 2024	Security, Network Management and Monitoring
July 2024	All About Data Management (Database management, Optimisation, Big Data, Data analysis, etc)
August 2024	Mobile/Web App Development, Optimisation and Security
September 2024	DevOps Special
October 2024	Cloud Special: Everything from management to implementation
November 2024	Containers and Managing Containers
December 2024	Open Source and Quantum Computing
January 2025	Open Source and IoT and Edge
February 2025	Open Source on Windows and Best in the world of Open Source (Tools and Services)